Experiment No. 10

Experiment Name-Internetworking with Routers in Cisco Packet Tracer

Aim

To study and implement internetworking using routers in Cisco Packet Tracer by connecting multiple computers and networks, configuring static IP addresses, establishing communication between different networks, configuring a wireless router with a DHCP server and Internet cloud, and connecting network devices using switches, routers, and Ethernet cables.

Procedure

Part A: Simple Internetwork Using a Router

1. Open Cisco Packet Tracer on the computer and create a new network topology.
2. Select the required devices from the device panel. Add one router, two switches, and four PCs to the workspace. Arrange the devices so that two PCs are connected to each switch, with the router connecting the two networks.
3. Connect PC0 and PC1 to the first switch using Copper Straight-Through Ethernet cables. Similarly, connect PC2 and PC3 to the second switch.
4. Connect the first switch to the router's GigabitEthernet0/0 interface and the second switch to the router's GigabitEthernet0/1 interface. Ensure that the correct interfaces are selected while making the connections.
5. Configure the IP addresses of the computers by opening each PC, selecting **Desktop → IP Configuration**, and entering the following details:
    - PC0: IP address 192.168.1.2, subnet mask 255.255.255.0, default gateway 192.168.1.1.
    - PC1: IP address 192.168.1.3, subnet mask 255.255.255.0, default gateway 192.168.1.1.
    - PC2: IP address 192.168.2.2, subnet mask 255.255.255.0, default gateway 192.168.2.1.
    - PC3: IP address 192.168.2.3, subnet mask 255.255.255.0, default gateway 192.168.2.1.

6. Configure the router interfaces. Open the router, select the CLI tab, and enter the following commands:
7. enable
8. configure terminal
9. interface gigabitEthernet 0/0
10. ip address 192.168.1.1 255.255.255.0
11. no shutdown
12. exit
13. interface gigabitEthernet 0/1
14. ip address 192.168.2.1 255.255.255.0
15. no shutdown
16. exit

end

17. Wait until the connected interfaces become active. Verify the IP configuration of each PC and ensure that its default gateway matches the router interface connected to its network.
18. Test communication between PCs in the same network. Open the Command Prompt of PC0 and execute ping 192.168.1.3. Similarly, test communication between PC2 and PC3 using ping 192.168.2.3.
19. Test communication between different networks by executing ping 192.168.2.2 from PC0. The packet should pass through the router to reach PC2. Repeat the test from another PC to verify communication in the reverse direction.
20. Select **Simulation Mode** in Cisco Packet Tracer and generate a Simple PDU between computers belonging to different networks. Observe how the packet travels from the source PC through the switch and router to the destination network. This helps demonstrate the role of the router in forwarding packets between different IP networks.

Part B: Wireless Router, DHCP Server and Internet Cloud

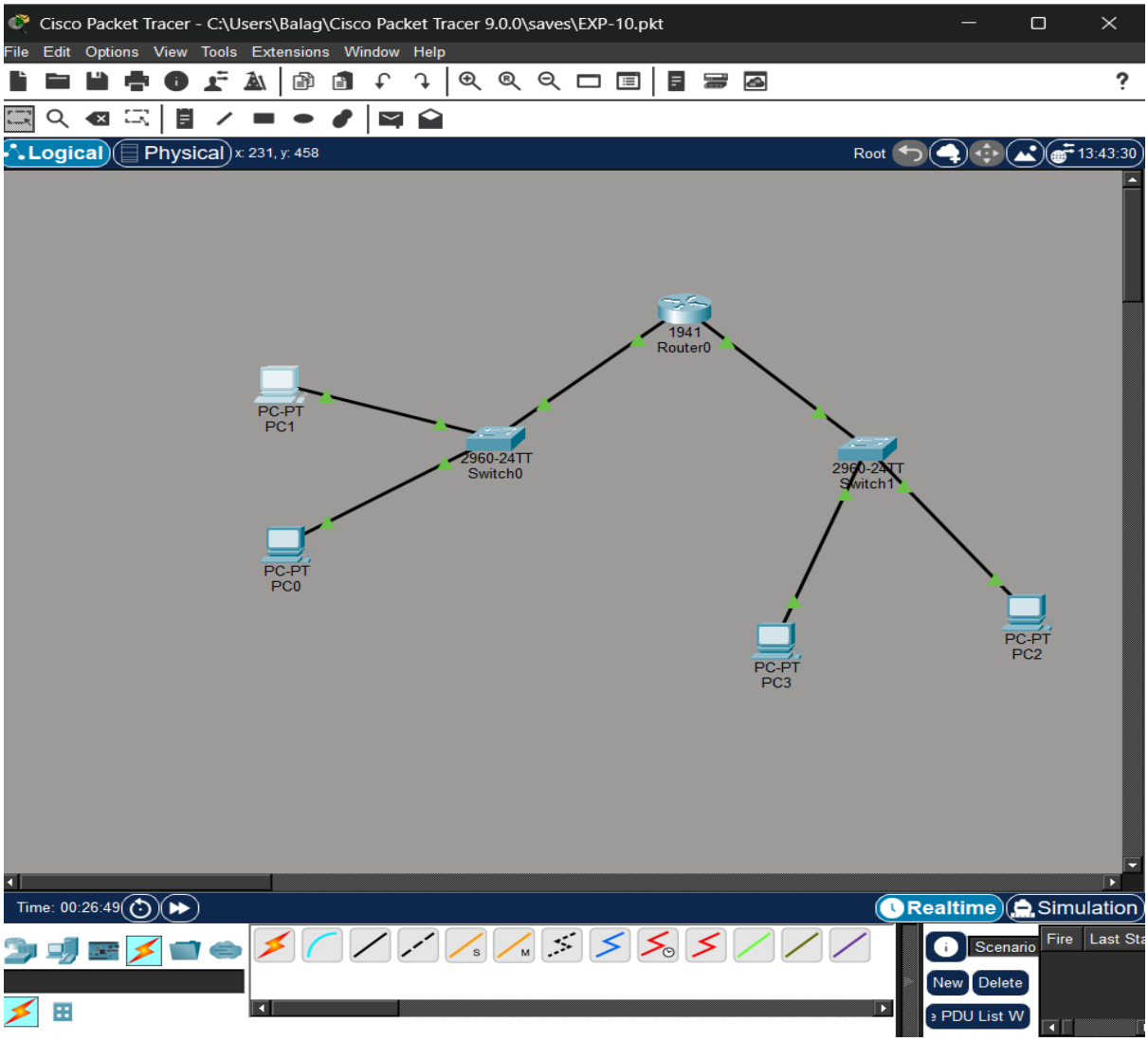
11. Create a new topology or extend the existing one by adding a wireless router, wireless-capable client devices, and an Internet cloud. Connect the appropriate devices according to the topology required for the experiment.

12. Open the wireless router's configuration window and configure the wireless network settings, including the wireless network name (SSID) and security settings if required. These settings allow compatible wireless devices to identify and connect to the network.
13. Configure the router's DHCP service, if available, to automatically assign IP addresses to client devices. Set the appropriate address range, subnet mask, and default gateway. Ensure that the DHCP address range does not conflict with any manually assigned addresses.
14. Connect the client devices to the wireless network or through the available Ethernet ports, depending on the device configuration. Select DHCP in the IP configuration settings of each client that should obtain its address automatically.
15. Verify that the clients receive valid IP addresses and network configuration details. Test connectivity between the client and its default gateway using the ping command. If the Internet cloud is configured with a functioning upstream network, test connectivity to the configured destination as well.

Part C: Inter-Network Connection Using Switches, Router and Ethernet Cables

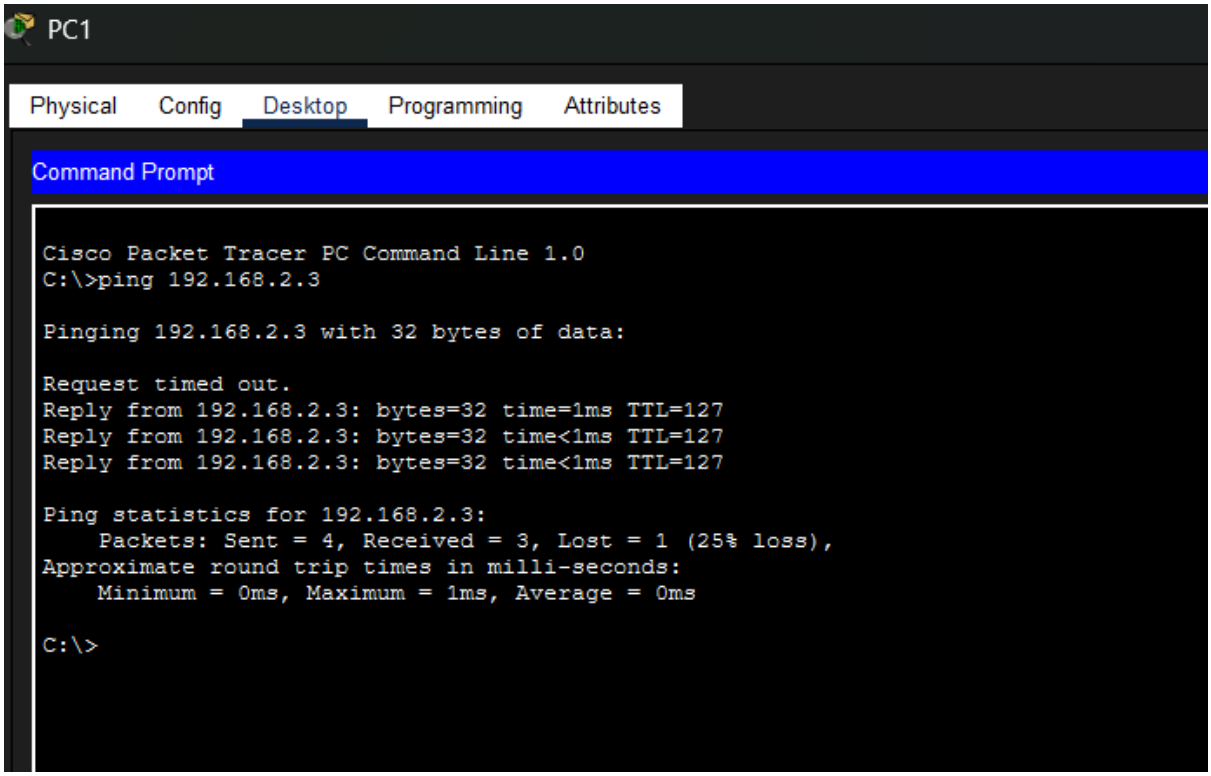
16. Arrange the network devices in the lab or in Cisco Packet Tracer according to the required topology. Use switches to connect computers within each local area network and a router to connect separate networks.
17. Connect the PCs to their respective switches and connect each switch to the appropriate router interface using Ethernet cables. Check the physical connections and interface status before testing communication.
18. Assign suitable IP addresses and subnet masks to all PCs. Configure each PC's default gateway using the IP address of the router interface in its local network. Ensure that devices within the same subnet have unique IP addresses.
19. Configure the router interfaces with IP addresses belonging to their respective subnets. Enable the interfaces and wait for the link status to become active. If the topology contains additional routers or remote networks, configure the routing information required for communication between them.
20. Use the ping command to verify connectivity between devices within the same network and across different networks. If communication fails, check the IP addresses, subnet masks, default gateways, cable connections, and router interface configuration.
21. Finally, use Simulation Mode to observe packet movement through the switches and router. Record whether communication succeeds and identify the devices through which the packets travel.

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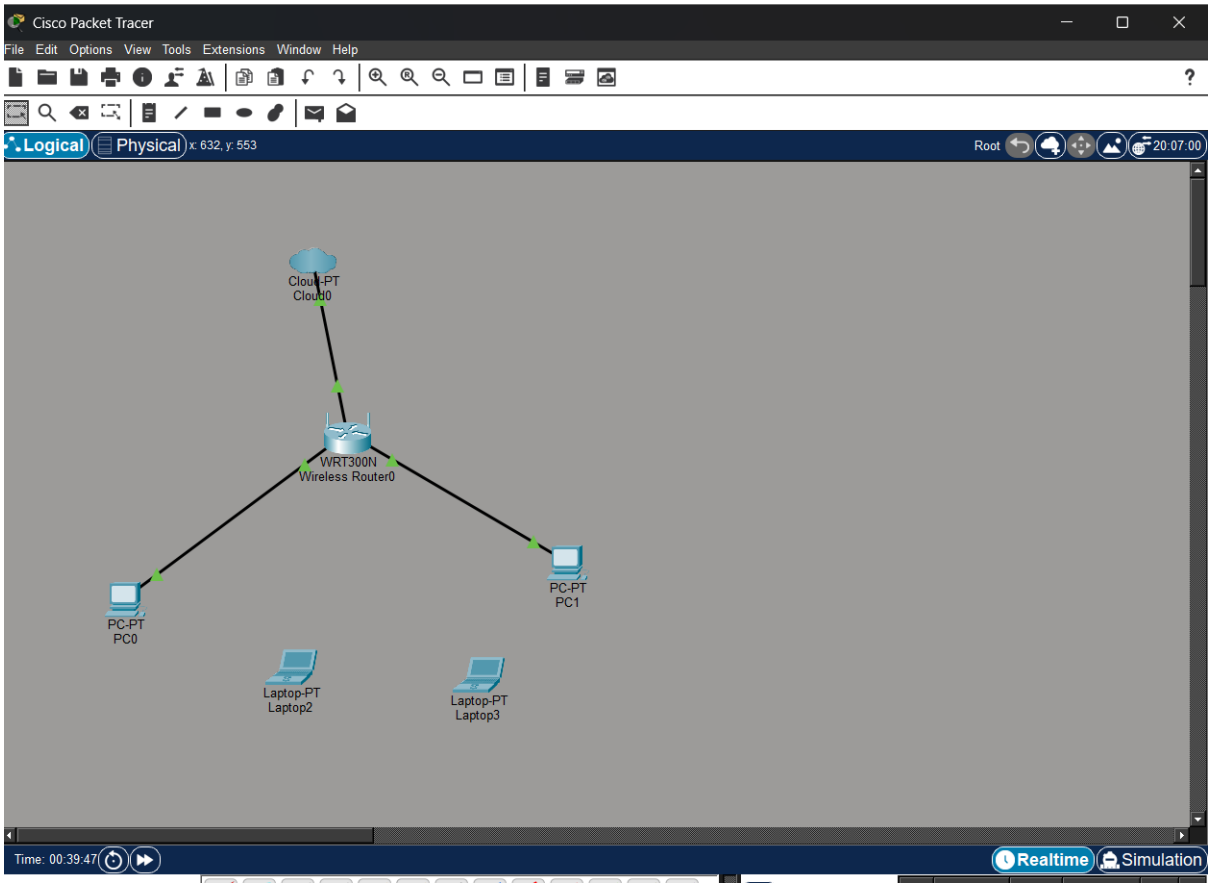

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
```

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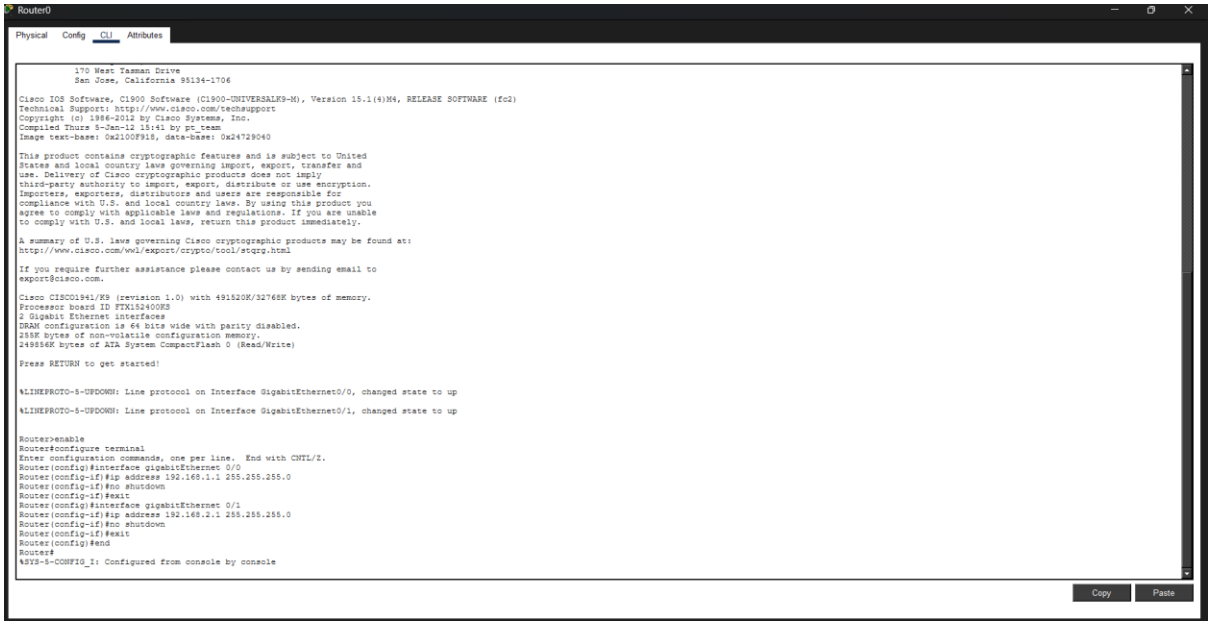
b) Design and configure an internetwork using wireless router DHCP server and internet cloud.

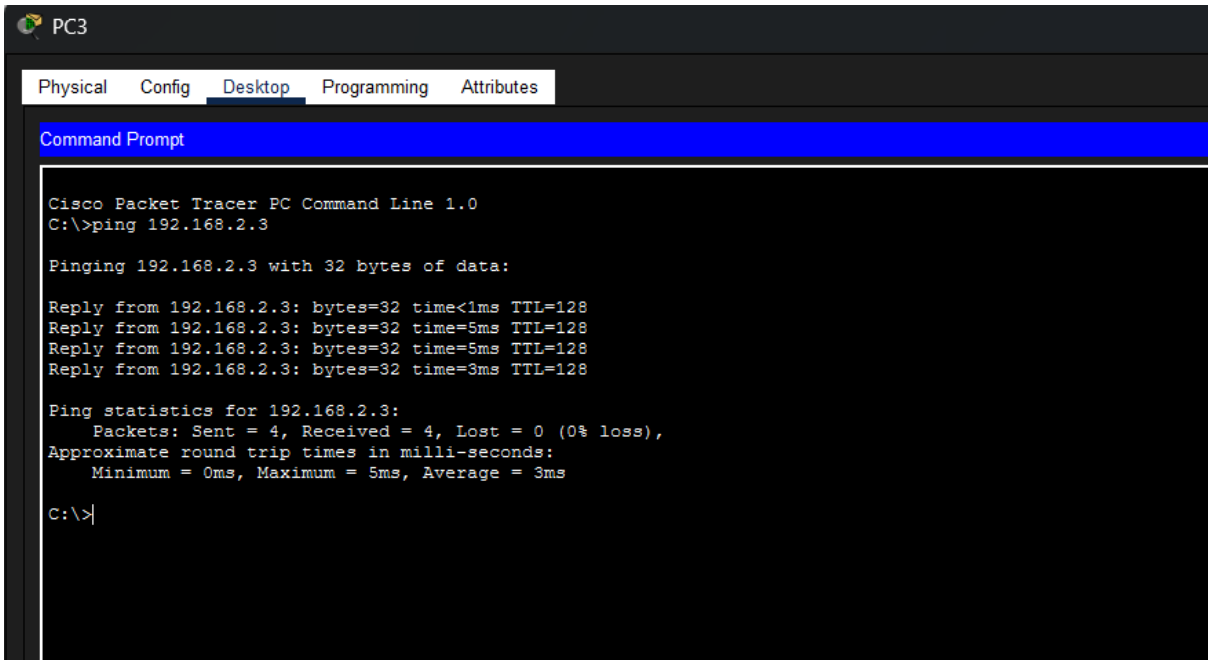
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c) Design and configure an inter-network in your lab using switch, router and Ethernet cables.

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Result

Thus, internetworking was studied and implemented using Cisco Packet Tracer. Multiple computers were connected through switches and a router, static IP addresses and default gateways were configured, and communication between different networks was tested using the ping command. The role of the router in forwarding packets between separate networks was observed using Simulation Mode. The configuration of a wireless router, DHCP address assignment, and the use of an Internet cloud were also studied as part of the experiment.